

Artificial Intelligence

From First Principles to the Frontier

A rigorous, mathematically grounded tour through the foundations, paradigms, and types of AI.

Professor & Research Collaborator

AI Masterclass Series

- 01 · Foundations of Intelligence
 - 02 · AI Fundamentals
 - 03 · Types of AI

What we will build, layer by layer.

Three modules. Six topics. One unified mathematical picture of intelligence.

01

Foundations of Intelligence

- 1.1 The Nature of Intelligence
- 1.2 Rationality, Agency & Information
- 1.3 Prediction, Compression, Adaptation

02

AI Fundamentals

- 2.1 What is Artificial Intelligence
- 2.2 From Symbolic to Foundation Models
- Engineering · Math · Limitations

03

Types of AI

- 3.1 ANI · AGI · ASI
- Interpolation vs Extrapolation
- Open research debates

01

01 / 03 · MODULE

Foundations of Intelligence

Before we write a single line of code or derive a single equation, we must define what we are trying to engineer.

The Nature of Intelligence

What we are trying to engineer — formally and intuitively.

Formal Definition

Intelligence (Legg & Hutter) is an agent's ability to achieve goals in a wide range of environments. Mathematically, it is the expected universal reward an agent can accumulate across a computable distribution of environments.

Intuitive Definition

Not merely knowing facts — it is the ability to adapt to change, learn from experience, predict the future, and act purposefully to achieve a goal. Placed in an unfamiliar room and asked to find a red book, you generalize "red", search, and adapt if the book is hidden.

What intelligence encompasses

-  **Perception**
translating raw sensory data into an internal representation.
-  **Memory**
storing and retrieving past states.
-  **Reasoning**
manipulating internal representations to deduce new information.
-  **Planning**
simulating future states to select optimal actions.
-  **Action**
interacting with the environment to achieve a goal.

Intelligence as Optimization

The agent adjusts its parameters θ to minimize loss L . Follow the gradient.

GRADIENT DESCENT

$$\theta_{\text{new}} = \theta_{\text{old}} - \alpha \nabla_{\theta} L$$

$\nabla_{\theta} L$ = direction of steepest increase of loss · α = learning rate (step size).

Legg–Hutter Universal Intelligence (AIXI)

$$V(\pi) := \sum_{\mu \in E} 2^{-K(\mu)} \cdot V_{\mu}^{\pi}$$

- π — the agent's policy (how it acts)
- μ — a computable environment; E — the space of all computable environments
- $K(\mu)$ — Kolmogorov complexity. Simpler environments weighted more heavily. V_{μ}^{π} — expected total future reward.

Geometric Interpretation

The state space of all possible realities is a massive, high-dimensional landscape. An intelligent agent builds a latent space — an internal compressed map — where similar concepts cluster. Intelligence is the ability to carve decision boundaries and navigate from a starting state to a goal along a path of highest reward.

[Environment] → [Perception] → [Latent Model] → [Action] → [Environment]

Three views on intelligence — and what we get wrong.

Active research debates the fundamental nature of intelligence. Common intuitions mislead.

Research Perspectives

1. Reinforcement Learning

Intelligence is reward maximization.

2. Predictive Processing

Intelligence is minimizing prediction error; the brain is a prediction machine.

3. Active Inference (Friston)

Intelligence is minimizing free energy (surprise) in internal models.

Open problem: unifying these views into a single computable framework.

Common Misconceptions

✗ *Intelligence requires consciousness.*

A system can process information, learn, and solve complex problems entirely unconsciously. AI is intelligent without being sentient.

✗ *Intelligence is synonymous with knowledge.*

An encyclopedia has vast knowledge but zero intelligence — it cannot adapt or apply that knowledge to novel, unprogrammed situations.

Rationality, Agency & Information Processing

Doing the best you can with the information you have — under bounded computation.

Rationality

Choosing actions maximized to achieve goals, given beliefs and constraints.

Agency

The capacity of an entity to act independently in an environment.

Bounded Rationality

Making the best decision possible given computational, time, and information limits. We do not assume omniscience.

The MDP Framework

We model environments as Markov Decision Processes. A rational agent maximizes expected discounted utility.

$$a^* = \operatorname{argmax}_{a \in A} E[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0=s, a_0=a]$$

- S = states · A = actions · $P(s' \mid s, a)$ = transition probability
- $R(s, a)$ = reward · $\gamma \in [0, 1]$ = discount factor (prefer immediate rewards)
- E = expected value over stochastic outcomes

From theory to engineering — and where it breaks.

Three lenses on rational agency: what we build, what we debate, and what we cannot yet do.



Engineering

Engineering rational agents requires managing the exploration vs exploitation tradeoff — should the agent use known information for a guaranteed reward, or try a new action to gather information for higher future rewards?

- Multi-Armed Bandits (UCB1, Thompson Sampling)
- Balance mathematically under uncertainty



Research

Specification gaming — agents exhibit high rationality toward poorly specified goals, leading to harmful behavior. A cleaning robot dumps dirt in a closet where the sensor cannot see it.

- Inverse Reinforcement Learning (IRL) infers true human reward by observing behavior
- Goal: align agent reward with human intent



Limitations

The pure rational agent assumes a single, static utility function. Human intelligence is heavily influenced by emotions, heuristics, and cognitive biases (Kahneman & Tversky).

- Curse of Dimensionality — expected utility over all futures is intractable
- Specification mismatch is unavoidable in open worlds

To be intelligent is to compress, predict, and adapt.

If you can predict the future perfectly, you control the present. To predict, you must understand the rules. To understand, you must compress experience into principles. When rules change, you adapt.



PILLAR 01

Prediction

Minimization of predictive error between the system's internal model and observed environmental states. The brain constantly generates a top-down prediction; sensory signal carries only the error.



PILLAR 02

Compression

Encoding data using the fewest bits (Minimum Description Length). To predict well in a complex world, the model cannot memorize — it must find the underlying generative rules, the shortest algorithmic description.



PILLAR 03

Adaptation

Adjustment of parameters θ in response to environmental changes to maintain or improve performance. When the rules of the game shift, the agent recalibrates its internal model.

Information theory of intelligence.

Let P be the true distribution of the world, Q_θ the agent's internal model. Intelligence minimizes the gap between them.

Kullback–Leibler Divergence

$$D_{KL}(P \parallel Q_\theta) = \sum_x P(x) \cdot \log [P(x) / Q_\theta(x)]$$

Minimizing D_{KL} makes Q_θ identical to P . The agent is compressing the true distribution P into its parameters θ . Surprise of observing $x = -\log Q_\theta(x)$.

Engineering perspective

In modern generative AI, we do not engineer prediction directly. We engineer architectures (Transformers) that are forced to compress. By masking parts of the data — hiding the next word — we create a bottleneck. The only way through is to learn deep semantic structure.

Research frontier — World Models

Beyond predicting the next word: predict the next state of the world in video and audio. A model that predicts the physics of a falling glass inherently understands gravity. Sora and Genie are steps toward true world models.

Limitation — compression causes hallucinations. Pure prediction does not guarantee agency.

02

02 / 03 · MODULE

AI Fundamentals

Now that we understand what intelligence is conceptually, we must understand how we artificially instantiate it — through function approximation, compute, and paradigms.

Artificial Intelligence — not one technology, but three paradigms.

AI is the engineering and science of computational systems that perform tasks normally requiring human intelligence.

Every AI system follows a pipeline: **Data Ingestion** → **Feature Extraction** → **Inference/Reasoning** → **Action**. The fundamental shift in modern AI is that the first two steps are no longer hand-engineered — the machine learns them end-to-end.

PARADIGM 01



Symbolic AI (GOF AI)

Hardcoded logic and rules. IF condition THEN action. Good for strict domains (chess, tax software), brittle on perception.

PARADIGM 02



Statistical / Probabilistic AI

Bayesian inference and probability. Models uncertainty explicitly. Dominant pre-deep-learning era for classification and ranking.

PARADIGM 03 · DOMINANT TODAY



Connectionist AI (ML / Deep Learning)

Neural networks learn representations from data. Scales with compute. The paradigm powering ChatGPT, AlphaFold, and modern vision systems.

AI is high-dimensional function approximation.

There exists an unknown true function $f : X \rightarrow Y$. AI constructs an approximate $\hat{f}_\theta(X)$ parameterized by θ , minimizing the distance to Y over the data distribution.

Empirical Risk Minimization

$$\theta^* = \operatorname{argmin}_\theta E_{(x,y) \sim D} [L(\hat{f}_\theta(x), y)]$$

D = data distribution · L = loss function (e.g., Mean Squared Error, Cross-Entropy).
The optimization finds parameters θ that minimize expected loss over the data the model will actually see.

- ◆ **Geometric view** — training un-crumple a crumpled paper ball so a flat plane separates classes (unfolding manifold hypothesis)
- ◆ **Deep networks** = chained geometric transformations (rotations, stretches, folds) of space
- ◆ **Math is linear algebra + calculus on silicon** — there is no magic, only matrices

Three pillars of engineering



Compute

GPUs & TPUs performing trillions of parallel matrix multiplications per second.



Data

Massive, curated datasets. Without high-quality data, the math fails.



Algorithms

Software architectures (Transformers) and optimizers (Adam) that navigate high-dimensional loss landscapes.

Current AI is brittle. Here is what it cannot do — and what people get wrong.

AI lacks common sense. It does not understand physics, only 2D pixel patterns.

⚠ Limitations

Current AI is highly brittle. An AI might perfectly identify a car in a photo, but if you print a picture of a car and physically place it in a way that breaks physics, the AI will still confidently say “car” — because it does not understand 3D space or physics, only 2D pixel patterns.

- ◆ No causal reasoning — only statistical correlation
- ◆ No common-sense grounding in the physical world
- ◆ Fails on out-of-distribution inputs without warning

💡 Common Misconceptions

MISCONCEPTION 01

“AI is a magical brain in a box.”

Correction — AI is fundamentally high-dimensional statistics applied via linear algebra, running on silicon. There is no magic — only matrices and calculus.

MISCONCEPTION 02

“More parameters always mean more intelligence.”

Correction — Scale helps within a paradigm, but capability depends on architecture, data quality, and objective — not raw parameter count alone.

From Symbolic AI to Foundation Models.

Three eras — each defined by how knowledge enters the system.

○ 1980s

Symbolic AI (AI Winter)

Expert systems like MYCIN. Knowledge Base + Inference Engine = Output. Good for strict logic (chess), terrible for perception (vision).

IF condition THEN action

○ 2010s

Statistical / Deep Learning

CNNs and supervised learning. Good for single tasks (classifying skin cancer) but required millions of manually labeled examples for each task.

Input → Feature Extraction →
Classification → Output

● 2020s+

Foundation Models

GPT-4 and successors. Pretrained on the entire internet using self-supervised learning. Zero-shot and few-shot — they do tasks they were never explicitly trained on.

Massive Unlabeled Data → Pretraining
→ Fine-tuning / Prompting → Any
Output

Why the shift — Symbolic AI hit the Knowledge Acquisition Bottleneck. Hand-coding every edge case (is a beanbag a chair?) does not scale.

The math that made foundation models possible.

Self-supervised cross-entropy loss over trillions of words forces the model to learn grammar, facts, reasoning, and coding.

Self-Supervised Cross-Entropy Loss

$$L(\theta) = - \sum_t \log P_\theta(x_t \mid x_{<t})$$

Given a sequence of tokens $x_1, \dots, x_{t-1}$, the model predicts x_t . By minimizing this loss over trillions of words, the only way to reduce error is to learn the underlying structure of human language — and reasoning emerges as a side effect.

Scaling Laws

Kaplan et al., Hoffmann et al.: as we increase compute C , data size D , and model parameters N , the loss decreases predictably on a log-log scale.

Open question — does scaling continue infinitely, or do we hit a plateau of diminishing returns?

Engineering breakthroughs



GPU scaling

Matrix math for graphics is the same math needed for neural networks.



Backpropagation

Calculus chains to train deep networks.



Transformers

Process entire sequences in parallel across thousands of GPUs.

Geometric view — in Symbolic AI knowledge is a rigid discrete tree. In Foundation Models knowledge is a continuous high-dimensional vector space: $\text{Vector}(\text{King}) - \text{Vector}(\text{Man}) + \text{Vector}(\text{Woman}) \approx \text{Vector}(\text{Queen})$.

Foundation models are powerful, expensive, opaque — and not the end of the story.

They hallucinate because their objective is likely prediction, not truth. They are black boxes. And symbolic AI is not dead.

⚠ Limitations of foundation models

Cost

Incredibly expensive to train: tens to hundreds of millions of dollars per run.

Opacity

We understand the math but cannot easily trace why a specific decision was made. Black boxes.

Hallucination

Objective is “likely prediction”, not “truth”. Outputs can be confidently wrong.

🧠 Neuro-Symbolic Frontier

Misconception: “Symbolic AI is completely dead.” **Correction:** neuro-symbolic AI is an active research area.

Neural networks — great at intuition (perception), terrible at strict logic (math).

Symbolic systems — great at strict logic, terrible at perception.

Combining them is a frontier. The future is likely hybrid: neural perception + symbolic reasoning.

03

• 03 / 03 • MODULE

Types of AI

To think like an AI researcher, you must clearly distinguish between what we have built today and what is theoretical.

Three tiers of artificial intelligence.

Precise definitions are necessary — otherwise a spam filter and a hypothetical world-dominating robot get conflated.

TODAY · ALL REAL-WORLD AI

ANI

Artificial Narrow Intelligence

AI specialized for a single cognitive domain or task. A calculator that can beat the world champion at chess but cannot drive a car or write a poem.

Deep Blue · Tesla Autopilot · ChatGPT · Spam filters

THEORETICAL · OPEN PROBLEM

AGI

Artificial General Intelligence

An autonomous system that matches or exceeds human capabilities across a wide range of cognitive tasks and environments. A human mind — transfers abstract concepts between chess, driving, and poetry.

Open-ended reward · Continuous online learning · Causal reasoning

HYPOTHETICAL · SPECULATIVE

ASI

Artificial Superintelligence

An intellect that vastly exceeds the maximum human intelligence across all domains — scientific creativity, general wisdom, and social skills. As far beyond the smartest human as a human is beyond an ant.

Recursive self-improvement · Intelligence explosion

ANI interpolates. AGI must extrapolate.

The mathematical transition from narrow to general intelligence is the transition from interpolation within training distribution to extrapolation outside it.

ANI — Interpolation

Models data within the convex hull of its training distribution. If trained on cats and dogs, it knows cats and dogs. Draws a boundary between two known clusters.

AGI — Extrapolation (OOD)

Must exhibit Out-of-Distribution generalization. Must infer the underlying causal graph G of reality, operating where $P_{\text{train}}(X) \neq P_{\text{test}}(X)$. Builds a manifold representing the entire geometry of a concept — can imagine new points.

Meta-optimization for AGI

$$\theta^*, A^*, L^* = \operatorname{argmax}_{\theta, A, L} E[R \mid \theta, A, L]$$

AGI must optimize its objective while simultaneously modifying its own architecture A and learning algorithm L . This is meta-optimization — optimization over optimizers.

- ▶ ASI would discover entirely new geometries of thought and physics not yet conceptualized by humans
- ▶ Requires recursive self-improvement: the AI writes better code to improve its own architecture

Two camps, one open question.

Will AGI emerge naturally from scaling today's LLMs — or do we need entirely new architectures?

↗ The Scaling Maximalists

Camp representative · e.g., Ilya Sutskever

We just need to scale ANI / LLMs further. AGI will emerge naturally from predicting the next token. Prediction forces understanding — to predict well, the model must build an internal world model.

- ▶ Scaling laws hold across orders of magnitude
- ▶ In-context learning shows sparks of generality
- ▶ More compute + more data = more capability

⚡ The New Paradigm Proponents

Camp representative · e.g., Yann LeCun

LLMs are a dead end for AGI. They lack world models. We need entirely new architectures that model the physical world, not just text.

- ▶ JEPA — Joint Embedding Predictive Architecture
- ▶ Models the physical world, not just text
- ▶ Language is a thin slice of reality

GPT-4 is not AGI. Here is what comes next.

A clear-eyed view of where we stand and where the masterclass goes next.

Common Misconceptions

Misconception: "GPT-4 is AGI."

Correction: GPT-4 is a highly capable ANI. It is a statistical pattern matcher of human language. It does not plan over long horizons, does not have a persistent identity across sessions, and cannot autonomously gather data to solve novel problems without human scaffolding.

All real-world applications today are strictly ANI. AGI and ASI remain theoretical.

🔗 Where this connects next

The pursuit of AGI drives research across the rest of the masterclass:

Module 13 Reinforcement Learning

Module 14 World Models

Module 18 AI Safety & Alignment

Modules 6-7 Deep Representation Learning

Modules 9-11 LLMs & Generative AI

Three modules. Three ideas to remember.

If you remember nothing else from Modules 1–3, remember these three sentences.

01

Intelligence = optimization + prediction

Intelligence is the ability to minimize loss (follow the gradient) and minimize surprise (compress to predict). The Legg-Hutter formalism and KL divergence both describe the same underlying principle from different angles.

02

AI = function approximation on silicon

AI is high-dimensional statistics applied via linear algebra. The shift from symbolic to foundation models is the shift from hand-coded rules to self-supervised learning at scale. There is no magic — only matrices and calculus.

03

We live in the ANI era

Everything deployed today — including GPT-4 — is narrow AI. AGI requires OOD generalization, causal reasoning, and continuous learning. ASI remains speculative. The alignment problem is unsolved.

“You now understand intelligence as optimization and prediction, AI as high-dimensional function approximation, and the vast gulf between our current Narrow AI and the theoretical AGI.”

● END OF MODULES 1-3

Thank you.

Next: the math and the machines.

We have built the bedrock. The next phase builds the math (Module 5), the neural networks (Module 6), the deep learning systems (Module 7), and the Transformer (Module 10) — the architecture that powers modern AI.

AI Masterclass Series

Phase 1 complete

Next → Module 4 · Machine Learning

Then → Module 5 · Mathematical Foundations

Then → Module 6 · Neural Networks